

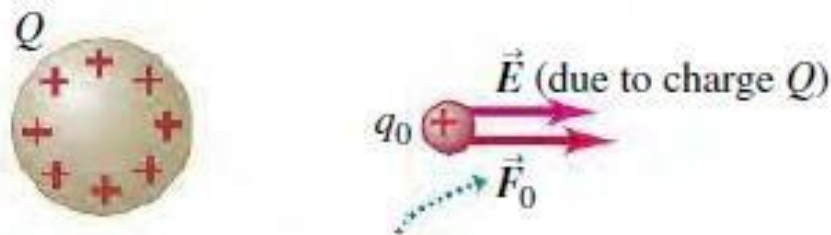
BS – 101 Engineering Physics

Lecture – 2

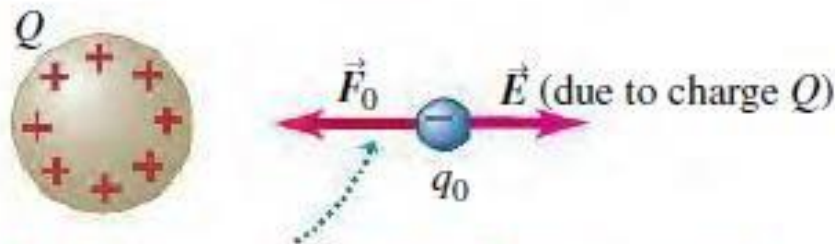
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Electric Field

- The electric force on a charged body is exerted by the electric field created by other charged bodies.



The force on a positive test charge q_0 points in the direction of the electric field.



The force on a negative test charge q_0 points opposite to the electric field.

Electric Field (Contd.)

- Electric Field is an Electric Force exerted on a test charge per unit test charge.

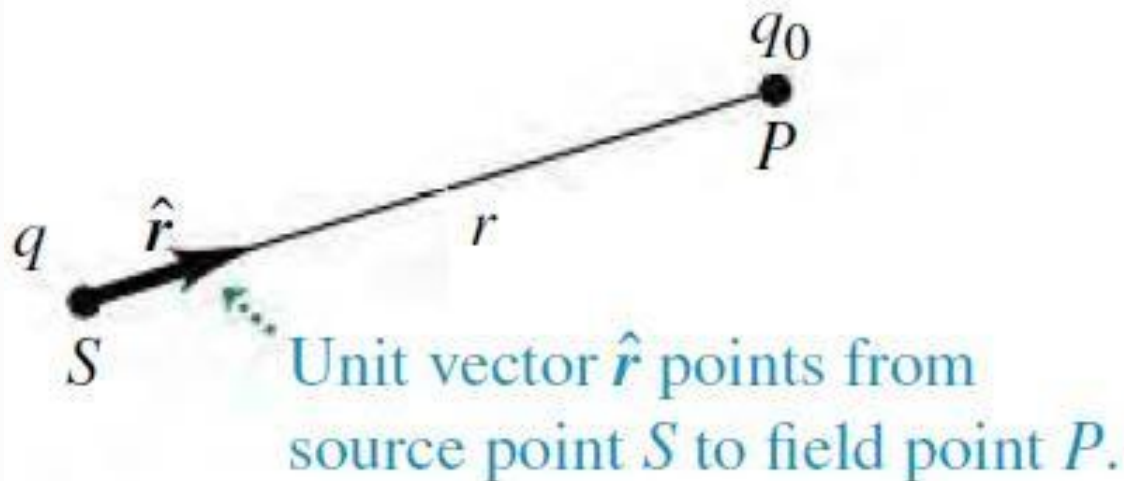
- $\vec{E} = \frac{\vec{F}_0}{q_0}$

- $\vec{F}_0 = q_0 \vec{E}$

- If charge is negative then the electric force will be opposite.
- If charge is positive then the electric force will be the same.

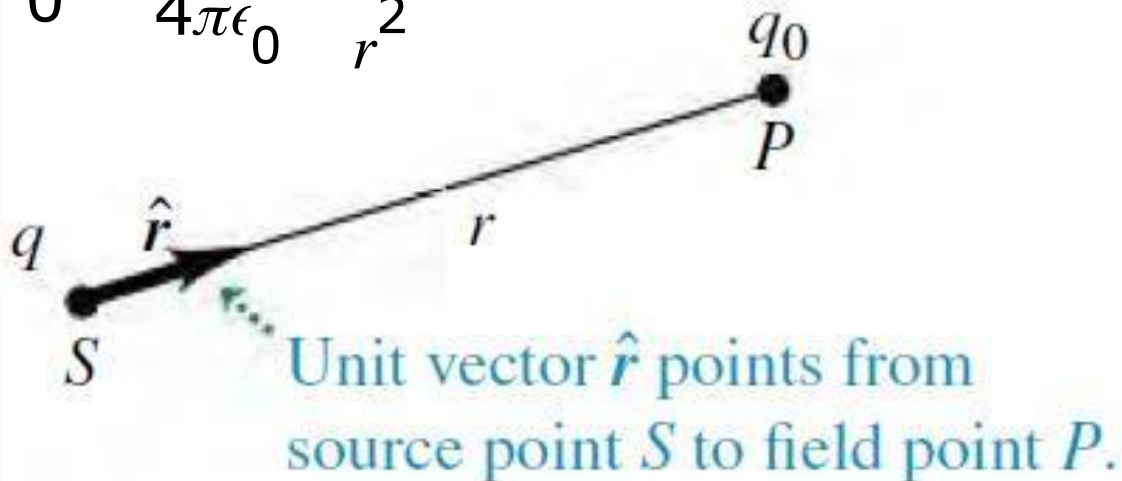
Electric Field of a Point Charge

- Suppose there is a point charge ' q ' which is at point S . There is also a test charge ' q_0 ' at point P .
- The electric force according to Coulomb's Law will be;



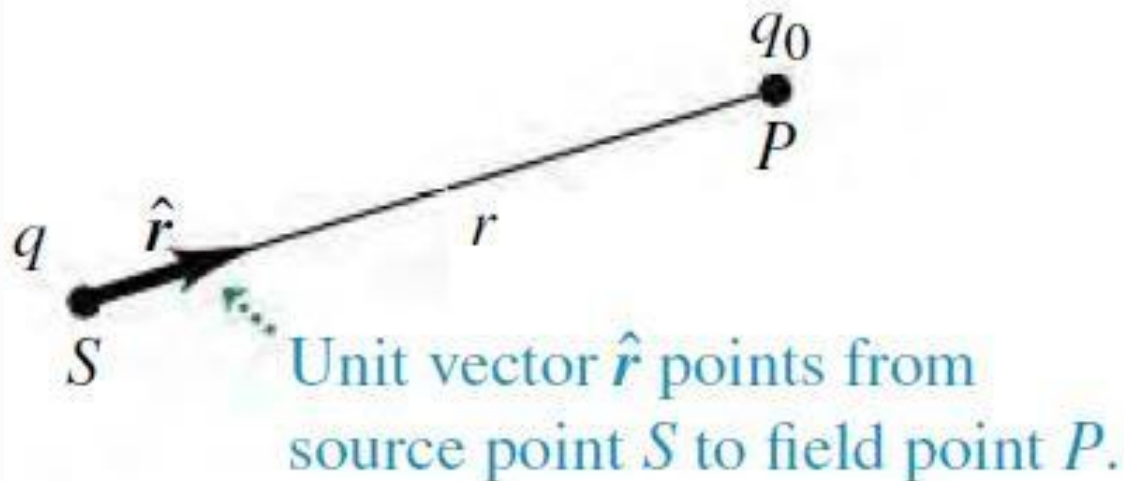
Electric Field of a Point Charge (Contd.)

- $F_0 = \frac{1}{4\pi\epsilon_0} \frac{q q_0}{r^2}$
- $\vec{F}_0 = q_0 \vec{E}$
- $q_0 E = F_0 = \frac{1}{4\pi\epsilon_0} \frac{q q_0}{r^2}$



Electric Field of a Point Charge (Contd.)

$$\begin{aligned} \bullet E &= \frac{1}{4\pi\epsilon_0} \frac{|q|}{r^2} \\ \bullet \vec{E} &= \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{r} \end{aligned}$$





Electric Field of a Point Charge (Unit)

- SI Units: Newton per Coulomb (N/C)



Electric Field of a Point Charge (Example)

- What is the magnitude of the electric field $\vec{E}$ at a field point 2.0 m from a point charge $q = 4.0 \text{ nC}$?



Electric Field of a Point Charge (Example)

- What is the magnitude of the electric field $\vec{E}$ at a field point 2.0 m from a point charge $q = 4.0 \text{ nC}$?
- 9 N/C



Electric Field of a Point Charge (Assignment

No. 1, Q.1)

- A point charge $q = -8.0 \text{ nC}$ is located at the origin. Find the electric-field vector at the field point $x = 1.2 \text{ m}$, $y = -1.6 \text{ m}$.



Electric Field of a Point Charge (Assignment No. 1, Q.2)

- Point charges $q_1 = +12 \text{ nC}$ and $q_2 = -12 \text{ nC}$ are 0.100 m apart. (Such pairs of point charges with equal magnitude and opposite sign are called *electric dipoles*.) Compute the electric field caused by q_1 , the field caused by q_2 , and the total field (a) at point a ; (b) at point b ; and (c) at point c .